Probability

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Example

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2. Tossing a coin
Outcome$= \{H, T\}$

Example

3. Tossing two coins
Outcome$= \{HH, HT, TH, TT\}$

Definition (Random experiment)

An experiment is **random** if its result can not be determined with certainty before hand.

Definition (Sample space)

The **sample space** S is a non-empty set that contains all possible outcomes of a random experiment.

Definition (Event)

Subset of a sample space

Definition (Null event and Certain event)

- An event, which does not contain any element called **null event or impossible event,** denoted by $\emptyset$
- If the event contains all the elements of the sample space, then it called a **certain event or sure event.**

Definition

- We say that an event has occurred if the outcome of the experiment belongs to the event.
- An event has not occurred if the outcome of the experiment is not in the event.

Definition (Mutually exclusive events)

- Two events A and B are **mutually exclusive** if both of them cannot occur simultaneously.
- That is, if the occurrence of one event prevents the occurrence of the other then they are mutually exclusive events.
- A and B are mutually exclusive if $A \cap B = \emptyset$.
- In tossing a coin, head and tail are mutually exclusive.
- In rolling a die, all six faces are mutually exclusive.

Definition (Independent events)

Events whose occurrence is not dependent on any other event.

Example

If we flip a coin in the air and get the outcome as head, then again if we flip the coin but this time, we get the outcome as tail. In both cases, the occurrence of both events is independent of each other.

Definition (Equally likely outcomes)

If all outcomes of a random experiment have equal chances of occurrence, then the outcome are said to be equally likely.

Example

- In tossing of an unbiased coin, head and tail are equally likely.
- In rolling a die, all six faces are equally likely.

Definition (Exhaustive cases)

The total number of possible outcomes of a random experiment is called exhaustive cases for that experiment.

Example

- In tossing a coin, exhaustive cases=2
- In tossing to two coins, exhaustive cases=4
- In tossing n coins, exhaustive cases= 2^n
- In rolling of a die, exhaustive cases=6

Definition (Favourable cases)

An outcome x is said to be favourable to an event A, if $x \in A$. The total number of outcomes favourable to A is called favourable cases to A.

Example

- In tossing to two coins, favourable cases for getting 2 heads is 1, for getting exactly one head is 2 and for getting at least two head is 3.
- In drawing a card from a pack, there are 4 cases favouring a king, 2 cases favouring a red queen and 26 cases favouring a black card.

Probability

Probability is a quantitative measure of chances of occurrence. There are 3 approaches to the study of probability.

1. Classical approach
2. Statistical or empirical approach
3. Axiomatic approach

Definition (Classical definition of probability)

If an even A can occur in m different ways out of a total of n ways all of which are equally likely and mutually exclusive, then the probability of the event A is given by

$$P(A) = \frac{m}{n} = \frac{\text{favourable cases}}{\text{exhaustive cases}}$$

- For a null set, $m = \emptyset$. Hence $P(\emptyset) = 0$
- For the sample space $m = n$. Hence $P(S) = 1$
- $0 \leq m \leq n$. Hence $0 \leq \frac{m}{n} \leq 1$. That is, $0 \leq P(A) \leq 1$
- m outcomes are favourable to $A \implies$ remaining $n - m$ outcomes are favourable to A'. Hence, $P(A') = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(A)$. That is, $P(A) + P(A') = 1$

Definition (Statistical definition of probability)

If an experiment is repeated several times under essentially homogeneous and identical conditions, then the limiting value of the ratio of the number of times the event occurs to the number of trials, as the number of trials become indefinitely large, is called the probability of that event.
That is, if an event A occurs m times in n trials then

$$P(A) = \lim_{n \to \infty} \frac{m}{n}$$

Axiomatic approach

1. For any event E, probability $P(E) \geq 0$.
2. When S is the sample space of an experiment, $P(S) = 1$.
3. If A and B are mutually exclusive outcomes, then $P(A \cup B) = P(A) + P(B)$.

Theorems

Theorem

$P(\emptyset) = 0$.

Theorem

If $\overline{A}$ *is the complementary event of* A*, then* $P(A) = 1 - P(\overline{A})$.

Theorem

If A *and* B *are any two events, then* $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Theorem

If $A \subset B$*, then* $P(A) \leq P(B)$.

Theorem

If A *and* B *are two independent events, then* $P(A \cap B) = P(A).P(B)$

Conditional Probability

- If we have two events from the same sample space, does the information about the occurrence of one of the events affect the probability of the other event?
- Let us try to answer this question by taking up a random experiment in which the outcomes are equally likely to occur.

- Consider the experiment of tossing three fair coins.
- The sample space of the experiment is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

- Since the coins are fair, we can assign the probability $1/8$ to each sample point.
- Let E be the event at least two heads appear and F be the event first coin shows tail.
- $E = \{HHH, HHT, HTH, THH\}$, $F = \{THH, THT, TTH, TTT\}$ and $E \cap F = \{THH\}$
- $P(E) = \frac{4}{8} = \frac{1}{2}$, $P(F) = \frac{4}{8} = \frac{1}{2}$ and $P(E \cap F) = \frac{1}{8}$

- Now, suppose we are given that the first coin shows tail, i.e. F occurs, then what is the probability of occurrence of E?

- Now, suppose we are given that the first coin shows tail, i.e. F occurs, then what is the probability of occurrence of E?
- With the information of occurrence of F, we are sure that the cases in which first coin does not result into a tail should not be considered while finding the probability of E.
- This information reduces our sample space from the set S to its subset F for the event E.
- In other words, the additional information really amounts to telling us that the situation may be considered as being that of a new random experiment for which the sample space consists of all those outcomes only which are favourable to the occurrence of the event F.

- Now, the sample point of F which is favourable to event E is THH.
- Thus, Probability of E considering F as the sample space $= \frac{1}{4}$ or Probability of E given that the event F has occurred $= \frac{1}{4}$
- This probability of the event E is called the conditional probability of E given that F has already occurred and is denoted by $P(E|F)$. Thus $P(E|F) = \frac{1}{4}$
- If E and F are two events associated with the same sample space of a random experiment, the conditional probability of the event E given that F has occurred, i.e. $P(E|F)$ is given by

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \text{ provided } P(F) \neq 0$$

Example

A die is tossed if the number is odd on the face, what is the probability that it is a prime?

Total Probability

Definition (Partition of a set)

The subsets of the sets $C_1, C_2, \ldots, C_n$ is said to be a partition of S, if

1. $\bigcup\limits_{i=1}^{n} C_i = S$
2. $C_i \cap C_j = \emptyset,\ i \neq j,\ \forall\ i, j$

Law of total probability

Let $C_1, C_2, \ldots, C_n$ form the partition of the sample space S, where all the events have a non-zero probability of occurrence. For any event A associated with S, according to the total probability theorem,

$$P(A) = \sum_{k=1}^{n} P(C_k)P(A|C_k)$$

Bayes theorem

Let $E_1, E_2, \ldots, E_n$ be a set of events associated with a sample space S, where all the events $E_1, E_2, \ldots, E_n$ have non-zero probability of occurrence and they form a partition of S. Let A be any event associated with S, then according to Bayes theorem,

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{k=1}^{n} P(E_k)P(A|E_k)}$$

Example

1. A person has undertaken a construction job. The probabilities are 0.65 that there will be strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

Example

2. suppose 3 companies x, y, z produce TVs. x produces twice as many as y while y and z produce same number. It is known that 2% of x, 2% of y, 4% of z are defected. All the TVs are produced are put into 1 shop and then 1 tv is chosen at random. What is the probability that the tv is defected. Suppose a tv chosen is defective, what is the probability that this tv is produced by company x.

Example

3. There are 3 boxes, the first one containing 1 white, 2 red and 3 black balls, the second one containing 2 white, 3 red and 1 black ball and the third one containing 3 white, 1 red and 2 black balls. A box is chosen at random and from it two balls are drawn at random. One ball is red and the other, white. What is the probability that they come from the second box?

Example

4. One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that he has the disease. The probability that he actually has the disease, is?

One-Dimensional Random Variables

Sample space is difficult to describe if elements are not numbers. Now we formulate results to represent sample space as numbers.

Definition (Random Variable)

Let S be a sample space of a random experiment. A function $f : S \to \mathcal{R}$ is called a random variable.

- Range(X) $= D$ which we assume to be
  1. Countable set
  2. Intervals in $\mathcal{R}$
- If D is countable, then X is discrete random variable. e.g., number of heads.
- If D is an interval in $\mathcal{R}$, then X is a continuous random variable. e.g., time, length.

Example

Consider the random experiment of tossing a die.

- Let $X : S \to \mathcal{R}$ be a function defined by $X(S) = 10S$
- Let $X : S \to \mathcal{R}$ be a function defined by $X(S) = \begin{cases} 0 & \text{if } S \text{ is odd} \\ 1 & \text{if } S \text{ is even} \end{cases}$

Probability Distributions

For discrete X: $P(X = x_i) = p_i$, $\sum p_i = 1$.
For continuous X: has density $f(x)$, $\int f(x)\,dx = 1$.

Example

1. X = number of heads in 3 coin tosses.

Example

2. Continuous case: $f(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Verify it's a pdf and find $P(X < 1/2)$.

Definition (Expectation)

$$E[X] = \begin{cases} \sum x_i p_i, & \text{if } X \text{ discrete,} \\ \int x f(x)\, dx, & \text{if } X \text{ continuous.} \end{cases}$$

Definition (Moment Generating Function (MGF))

$$M_X(t) = E[e^{tX}] = \begin{cases} \sum e^{tx_i} p_i, & \text{if } X \text{ is discrete} \\ \int e^{tx} f(x)\, dx. & \text{if } X \text{ is continuous} \end{cases}$$

Moments: $E[X^r] = M_X^{(r)}(0)$.

Example

1. Discrete: $P(X = 0) = P(X = 1) = 1/2$.

Example

2. X with pdf $f(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Definition (Probability mass function)

If X is discrete random variable which takes the values $x_1, x_2, x_3, \dots$ such that $P(X = x_i) = p_i$, then p_i is called probability mass function if

1. $p_i \geq 0$ for all i
2. $\sum_{i=1}^{n} p_i = 1$

Definition (Probability density function)

A function is probability density function if

1. $f(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

Definition (Distribution function)

The distribution function (also called the cumulative distribution function, or CDF) of a random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

That means $F(x)$ gives the total probability accumulated up to the point x.

Properties of distribution function

- $F(x)$ is non-decreasing.
- $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$
- $P(X > x) = 1 - F(x)$

1. If X is a random variable with pdf

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Find distribution function.

2. If X is a random variable with pmf $p(x) = \begin{cases} \frac{x}{15}, & x = 1, 2, 3, 4, 5 \\ 0, & \text{otherwise} \end{cases}$

Find

(a) P(X=1 or X=2)

(b) $P(\frac{1}{2} < x < \frac{5}{2} | x > 1)$

3. If X is a crv with pdf $f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$

Find

(a) c

(b) $P(X > 1)$

(c) distribution function.

4. The diameter of an electric cable is assumed to be a crv with pdf

$$f(x) = \begin{cases} kx^3(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find

(a) k

(b) $P(X < \frac{2}{3})$

(c) distribution function.

5. A crv X has the pdf $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

Find

(a) $P(X < \frac{1}{2})$

(b) $P(X > \frac{3}{4} | X > \frac{1}{2})$

(c) $P(X < \frac{3}{4} | X > \frac{1}{2})$

Two-Dimensional Random Variables

Definition

Let S be the sample space associated with a random experiment. Let $X = X(s)$ and $Y = Y(s)$ be two functions assigning a real number to each outcomes $s \in S$. We call (X, Y), a two-dimensional random variable.

Definition (Discrete)

(X, Y) is a two dimensional discrete random variable if the possible values of (X, Y) are finite or countably infinite.

Definition (Continuous)

(X, Y) is two dimensional continuous random variable if (X, Y) can assume all values in some non-countable set of Euclidean plane.

Definition (Joint Probability Distribution)

1. **For discrete (X, Y):** For each possible outcome (x_i, y_j), we associate a number $p(x_i, y_j)$ representing $P(X = x_i, Y = y_j)$ and satisfying the following conditions.

- (i) $p(x_i, y_j) \geq 0$ for all (x, y)
- (ii) $\sum_i \sum_j p(x_i, y_j) = 1$

The function p defined for all (x_i, y_j) in the range space of (X, Y) is called the probability function of (X, Y).

2. **For continuous (X, Y):** The joint probability function f is a function satisfying the following conditions.

- (i) $f(x, y) \geq 0$ for all $(x, y) \in R$
- (ii) $\int \int f(x, y) dxdy = 1$

Definition (Distribution function)

Let (X, Y) be a two-dimensional random variable. The distribution function or cumulative distribution function F of the two-dimensional random variable (X, Y) is $F(X, Y) = P(X \leq x, Y \leq y)$.

Definition (Marginal Probabilities)

The **marginal distributions** describe the individual behavior of X or Y.

Discrete:

$$p_X(x) = \sum_y p(x, y), \quad p_Y(y) = \sum_x p(x, y)$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

$X \setminus Y$	1	2
0	0.1	0.3
1	0.2	0.4

$$\sum_x \sum_y p(x, y) = 0.1 + 0.3 + 0.2 + 0.4 = 1$$

$$p_X(0) = 0.1 + 0.3 = 0.4, \quad p_X(1) = 0.2 + 0.4 = 0.6$$

$$p_Y(1) = 0.1 + 0.2 = 0.3, \quad p_Y(2) = 0.3 + 0.4 = 0.7$$

Hence,

$$p_X(x) : (0, 0.4), (1, 0.6), \quad p_Y(y) : (1, 0.3), (2, 0.7)$$

Conditional Distributions

Definition

Conditional probabilities describe the distribution of one variable given that the other is fixed.

Discrete:

$$P(X = x|Y = y) = \frac{p(x,y)}{p_Y(y)}, \quad P(Y = y|X = x) = \frac{p(x,y)}{p_X(x)}$$

Continuous:

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}, \quad f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

Summary

Concept	Discrete	Continuous
Joint distribution	$p(x,y)$	$f(x,y)$
Marginal of X	$\sum_y p(x,y)$	$\int f(x,y)\,dy$
Marginal of Y	$\sum_x p(x,y)$	$\int f(x,y)\,dx$
Conditional of $X \mid Y = y$	$\dfrac{p(x,y)}{p_Y(y)}$	$\dfrac{f(x,y)}{f_Y(y)}$
Conditional of $Y \mid X = x$	$\dfrac{p(x,y)}{p_X(x)}$	$\dfrac{f(x,y)}{f_X(x)}$

If $f(x,y) = f_X(x)f_Y(y)$, then X and Y are independent.

1. Suppose that 2 cards are drawn at random from a pack of cards. Let X be the number of aces and Y be the number of queens obtained.

a. Find joint probability distribution of (X, y)
b. Find marginal pdf of X and Y
c. Find $P(X \leq 1, Y > 1)$
d. Find $P(X + Y > 1)$
e. Find conditional probability distribution function of X given $Y = 1$
f. Find conditional probability distribution function of Y given $X = x$

2. Suppose that the 2-dimensional continuous random variable (X, Y) has joint pdf given by

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{Otherwise} \end{cases}$$

Find

a. $P(X + Y \geq 1)$
b. Marginal pdf of X and Y
c. $P(X > \frac{1}{2})$
d. $P(Y < X)$
e. $P(Y < \frac{1}{2} | X < \frac{1}{2})$

Definition (Expectation and Variance)

$$E[X] = \sum_i x_i p_X(x_i), \quad E[Y] = \sum_j y_j p_Y(y_j)$$

$$Var(X) = E[X^2] - (E[X])^2.$$

Definition (Covariance and Correlation Coefficient)

$$Cov(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}.$$

Interpretation of Correlation Coefficient (ρ)

- The correlation coefficient measures the **strength and direction** of the linear relationship between two random variables X and Y.

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} \quad \text{where} \quad -1 \leq \rho_{XY} \leq 1$$

General Conclusions

- $\rho = +1$: Perfect positive linear relationship.
- $\rho = -1$: Perfect negative linear relationship.
- $\rho = 0$: No linear relationship (variables may still be related nonlinearly).
- $|\rho| < 0.3$: Weak correlation.
- $0.3 \leq |\rho| < 0.7$: Moderate correlation.
- $|\rho| \geq 0.7$: Strong correlation.

Example

1. p_{XY} defined by

$X \backslash Y$	0	1
0	0.1	0.2
1	0.3	0.4

Find

1. Expectations of X and Y.
2. Variation of X and Variation of Y
3. Standard deviation of X and Standard deviation of Y
4. Covariance
5. Correlation coefficient

Definition (Generating function)

Let $a_1, a_2, \ldots, a_n, \ldots$ be a sequence of real numbers. Using real variables s, we define a function,

$$A(s) = a_0 + a_1 s + a_2 s^2 + \ldots = \sum_{k=0}^{\infty} a_k s^k$$

If this series is convergent in some interval $-s_0 < s < s_0$, then $A(s)$ is called generating function of sequence $a_0, a_1, a_2, \ldots$

Definition (Probability generating function)

Suppose that X is a random variable which assumes non-negative integral values, $0, 1, 2, \ldots$ and $P(X = x) = p_x$, $k = 0, 1, 2, \ldots$ and $\sum\limits_{x=0}^{\infty} p_x = 1$. Let

$$p(s) = \sum_{x=0}^{\infty} p_x s^x = E[s^x]$$

Remark:

- Differentiate $p(s)$ with respect to s

$$p'(s) = \sum_{x=1}^{\infty} x p_x s^{x-1}, \ -1 < s < 1$$

Put s=1,

$$p'(1) = \sum_{x=1}^{\infty} x p_x = \sum_{x=1}^{\infty} x P(X) = E[X]$$

- Differentiate $p''(s)$ with respect to s

$$p''(s) = \sum_{x=1}^{\infty} x(x-1)p_x s^{x-2}, \ -1 < s < 1$$

Put s=1,

$$p''(1) = \sum_{x=1}^{\infty} x(x-1)p_x = \sum_{x=1}^{\infty} x(x-1)P(X) = E[X(X-1)]$$

- $E[X^2] = E[X(X-1) + X] + E[X(X-1)] + E[X] = p''(1) + (p'(1))$
- $V(x) = E[X^2] - (E[X])^2 = p''(1) + p'(1) - (p'(1))^1$
- Therefore, $E[x] = p'(1)$ and $V(x) = p''(1) + p'(1) - (p'(1))^2$

Specific Random Variables

Discrete:

1. Bernoulli
2. Binomial
3. Poisson
4. Geometric

Continuous:

1. Exponential
2. Normal

Bernoulli distribution

- A random variable X is said to be Bernoulli distributed, if x takes the values 1 and 0 with $P(X=1)=p$ and $P(X=0)=q=1-p$
- Probability mass function of Bernoulli's distribution is $p(x)=p^x(1-p)^{1-x}$
- **Mean or Expectation:**

$$\mu = E[X] = \sum xp(X) = 1.P(X=1)+0.P(X=0) = 1.p+0.q = p$$

- $E[X^2] = \sum X^2 p(X) = 1^2.p + 0^2.q = p$
- **Variance:** $V(X) = E[X^2]-\mu^2 = p-p^2 = p(1-p) = pq$

Example

1. A machine produces items and each item is defective with probability 0.1 and non-defective with probability 0.9.

- Define Bernoulli's r.v. X for item is defective.
- Find pmf, expectation and variance.

Example

2. A student guesses the answer to a true or false question.

- Define Bernoulli's r.v. X for answer is correct.
- Find pmf, expectation and variance.

Binomial Distribution

Definition

A random variable X is said to be binomial distributed with parameters n and p if X takes the values $0, 1, 2, \ldots, n$ with
If X = number of successes in n independent Bernoulli trials with success prob. p, then

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \ldots, n, \text{ and } p + q = 1.$$

Here

- n is number of trials
- p is success
- q=1-p
- x is Number of success in n trials

PMF, Mean and Variance of Binomial distribution

Probability mass function of binomial distribution is

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & \text{if } x = 0, 1, \ldots, n \\ 0, & \text{Otherwise} \end{cases}$$

The pgf of X is

$$\begin{aligned} p(s) &= \sum_{x=0}^{\infty} p_x s^x \\ &= \sum_{x=0}^{n} P(X = x) s^x \\ &= \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x} s^x \\ &= (q + ps)^n \end{aligned}$$

Therefore, $p(s) = (q + ps)^n$.

Differentiate with respect to s.
$p'(s) = n(q + ps)^{n-1}.p$
Put $s = 1$, $p'(1) = n(q + p)^{n-1}.p = n(1 - p + p)^{n-1}.p = np$
This implies **Expectation or mean:** $E[X] = p'(1) = np$
$p''(s) = n(n-1)(q + ps)^{n-2}p^2$
Put $s = 1$, $p''(1) = n(n-1)p^2$

$$\begin{aligned} V(X) = E[X^2] - (E[x])^2 &= p''(1) - p'(1) - (p'(1))^2 \\ &= n(n-1)p^2 + np - n^2p^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np(1-p) \\ &= npq \end{aligned}$$

Therefore, **Variance:** $V(X) = npq$

Example

1. Disks produced by a company will be defective with probability 0.01. The company sells the disks in a packet of 10 and offers money back guarantee if more than one defective disk found in a packet. What is the proportion of packets returned? If someone buys 3 packets, what is the probability that exactly one of them should be replaced?

Example

2. Numbers are selected at random one at a time from the 2-digit numbers 00,01,02,...,99, with replacement. An event A occurs if and only if the product of the two digits of a selected number is 18. If 4 numbers are selected, find the probability that the event A occurs at least 3 times.

What is Normal Distribution?

- The Normal Distribution (Gaussian Distribution) is a continuous probability distribution that describes how the values of a variable are distributed.
- Most of the observations cluster around the mean.
- The graph is bell-shaped and symmetric about the mean.
- Denoted as $X \sim N(\mu, \sigma^2)$

Probability Density Function (PDF)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

- μ: Mean (center of distribution)
- σ: Standard deviation (spread)
- Total area under the curve $= 1$
- Symmetric about $x = \mu$

Properties

- If $X_1, X_2, \ldots, X_n$ are independent random variables that are normally distributed, then sum S is also normally distributed. Also,

$$S = X_1 + X_2 + \cdots + X_n$$

$$E[S] = E[X_1] + E[X_2] + \cdots + E[X_n]$$
$$V[S] = V[X_1] + V[X_2] + \cdots + V[X_n]$$

- If $X_1, X_2, \ldots, X_n$ are independent random variables that are normally distributed and the random variable is defined as $Q = a_1X_1 + a_2X_2 + \cdots + a_nX_n + b$. Then Q is also normally distributed. Also,

$$E[Q] = a_1E[X_1] + a_2E[X_2] + \cdots + a_nE[X_n] + b$$

$$V[Q] = a_1^2V[X_1] + a_2^2[X_2] + \cdots + a_n^2V[X_n]$$

Graphical Features

- Bell-shaped and symmetric.
- Mean = Median = Mode.
- $x-$ axis: random variable $y-$ axis: probability density
- Tails approach, but never touch, the x-axis.
- The spread is controlled by σ.

Example

1. The weight of a module used in a spacecraft is to be closely controlled. Since the module uses a bold-nut-washer assembling in numerous places. A study was conducted to find the distribution of weights of these parts. It was found that the three weights in grams are normally distributed with the following means and variances.

	means	variances
Bolt	312.8	2.67
Nut	53.2	0.85
Washer	17.5	0.21

Find the distribution of the weight of the assembly. Find mean, variance and standard deviation.

Example

2. The four independent normal random variable X_1, X_2, X_3, X_4 have the following means and variances.

	means	variances
X_1	12	4
X_2	-5	2
X_3	8	5
X_4	10	1

Find mean and variance of $Q = X_1 - 2X_2 + 3X_3 - 4X_4 + 5$.
Also find standard deviation of Q.